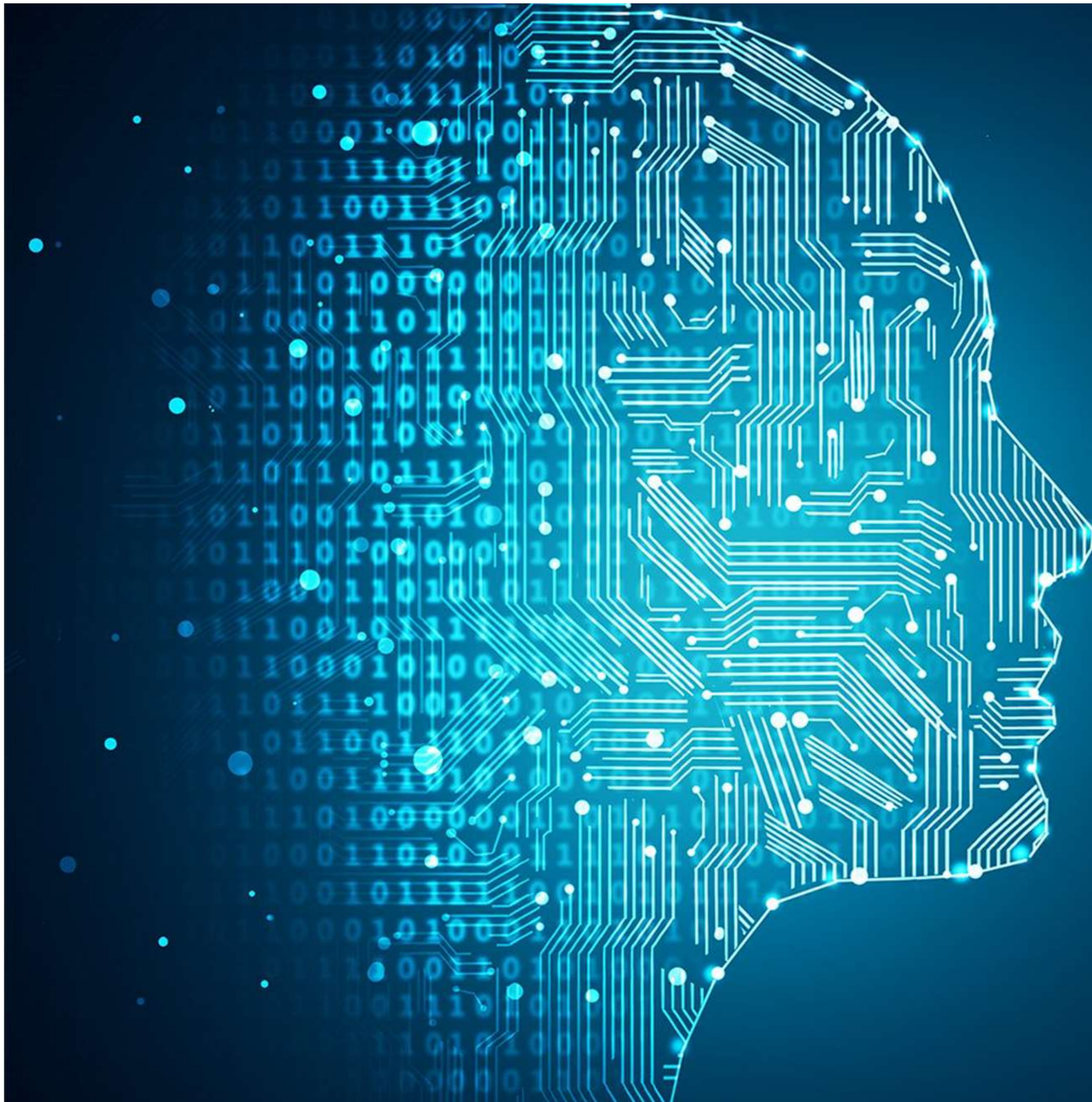


Welcome to

AASD 4014 – DL II

Credit hours: 42

Instructor: Dr. Moe Fadaee

Email: mo.fadaee@georgebrown.ca

George Brown College

Image Segmentation

Image Segmentation: Applications

1. Film industry and green screen
2. Self driving cars
3. Health and Medical imaging
4. Face detection
5. Security
6. Geology
7. Agriculture
8. Image compression, editing...
9. …

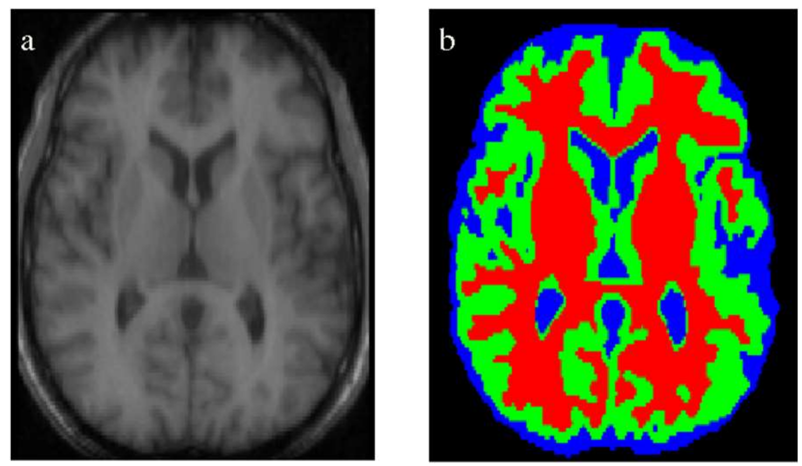


Image Segmentation: Different types

Classification and segmentation

1. Image classification:

   Given a set of known labels this task refer to classifying an input image to one of those predefined labels, so the only output for an input image would be a label.

   This is one of the most fundamental tasks in computer vision.

2. Classification with localization

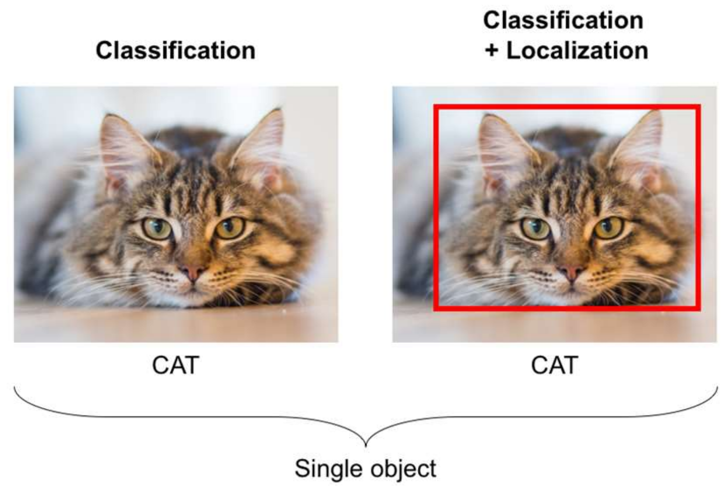
   In localization the computer along with the label, should also detect the location of the object (with the bounding box)

   However it is still one class of objects

3. Object Detection:

   In object detection we expect the computer to detect the location of multiple classes of objects (with bounding box).

   This is same as localization except this is for several classes of objects.



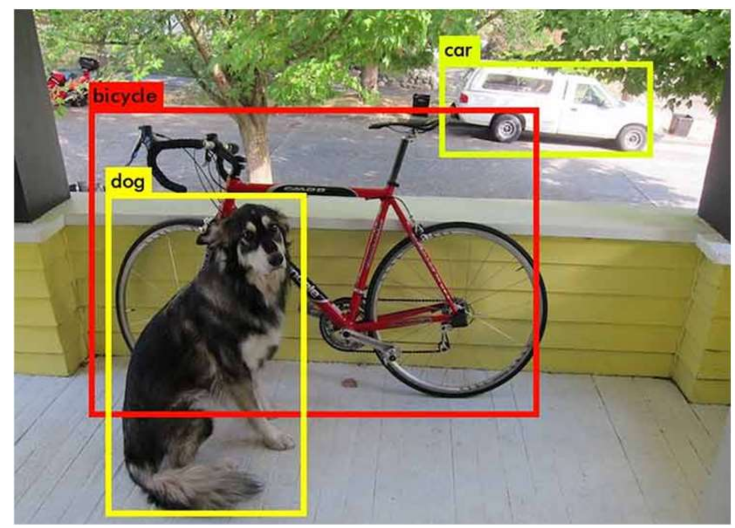


Image Segmentation: Different types

Classification and segmentation

4. Semantic Segmentation

   Semantic Segmentation is basically the task of classifying every single pixel of an image. This is clustering pixels into different categories. Semantic Segmentation is also referred to as dense prediction.

   Here the output is also an image with typically the same size where each pixel belong to class.

Input

1: Person
2: Purse
3: Plants/Grass
4: Sidewalk
5: Building/Structures

3	3	3	3	3	3	3	3	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	3	3	3	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	3	1	1	3	3	3	3	5	5	5	5	5	5
3	3	3	3	3	1	1	1	1	3	3	3	5	5	5	5	5	5
3	3	3	3	3	3	1	1	3	3	3	5	5	5	5	5	5	5
5	5	3	3	3	3	1	1	3	3	5	5	5	5	5	5	5	5
4	4	3	4	1	1	1	1	1	1	4	4	4	5	5	5	5	5
4	4	3	4	1	1	1	1	1	1	4	4	4	4	4	5	5	5
4	4	4	1	1	1	1	1	1	1	1	4	4	4	4	4	4	4
3	3	3	1	1	1	1	1	1	1	1	4	4	4	4	4	4	4
3	3	3	1	2	2	1	1	1	1	1	4	4	4	4	4	4	4
3	3	3	1	2	2	1	1	1	1	1	4	4	4	4	4	4	4

Semantic Labels

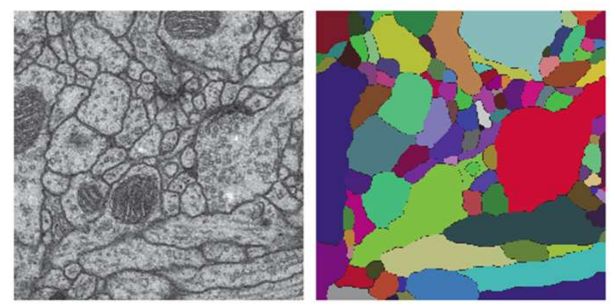


Image Segmentation: Different types

Classification and segmentation

5. Instance Segmentation

   This is similar to semantic segmentation other than in this approach the model will distinguish different instances of each class (semantic segmentation does not, i.e all instances of class car are classify to be part of the same class).

   As an example look at the chairs in this image:

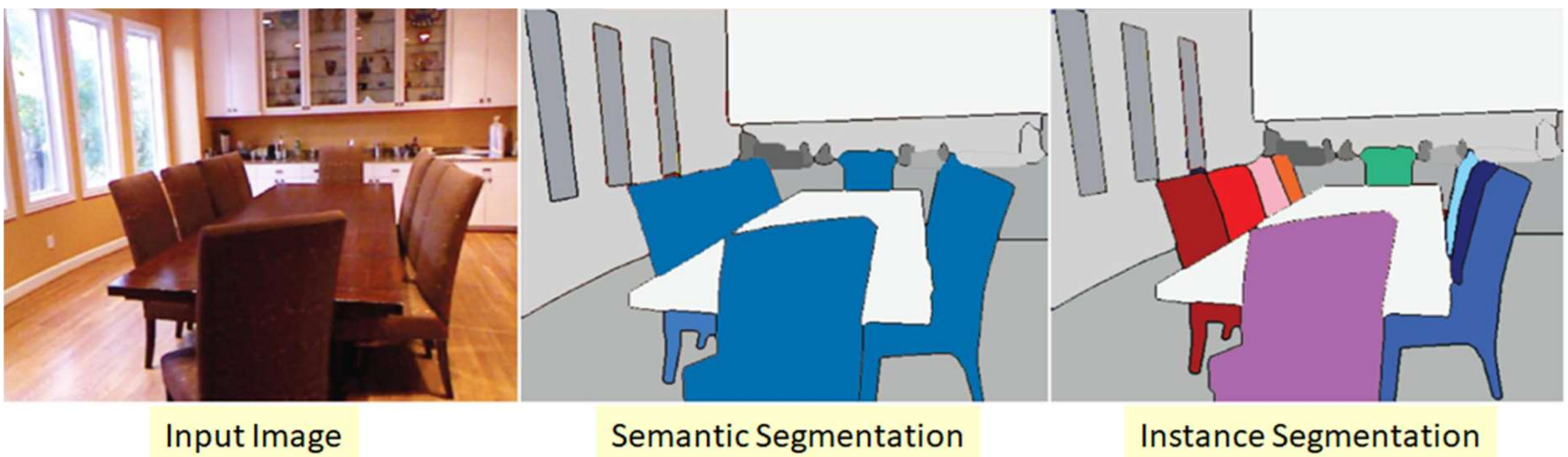


Image Segmentation: Methodologies

Here are different methods that one can employ to perform image segmentation:

1. Thresholding Methods

   This is a very simple way for image segmentation and works based on a threshold.

   1. Global thresholding: based on any appropriate value:

$$q(x,y) = \begin{Bmatrix} 1, if\ p(x,y) > T \\ 0, if\ p(x,y) \leq T \end{Bmatrix}$$

   1. Variable Thresholding: in this approach the value of T can vary over the image, this approach is divided into two methods:
      A. Local threshold: T depend on the neighborhood of x and y
      B. Adaptive threshold: T is a function of x and y

   1. Multiple Threshold:

$$\mathrm{q}(x,y) = \begin{Bmatrix} m, if\ p(x,y) > T1 \\ n, if\ p(x,y) \leq T1 \\ o, if\ p(x,y) \leq T0 \end{Bmatrix}$$

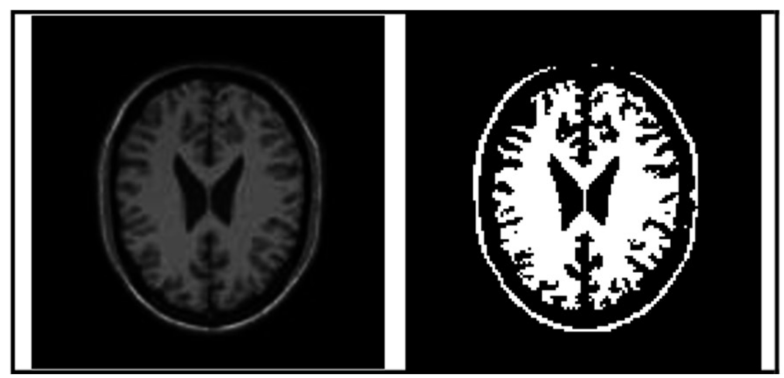


Image Segmentation: Methodologies

Here are different methods that one can employ to perform image segmentation:

2. Edge Based Methods (Boundary based approach)

   This method focus on detecting rapid change of intensity value in an image because a single intensity value does not provide good information about edges. Edge detection techniques locate the edges where either the first derivative of intensity is greater than a particular threshold or the second derivative has zero crossings.

   Edge-based segmentation algorithms work to detect edges in an image, based on various discontinuities in grey level, colour, texture, brightness, saturation, contrast etc. To further enhance the results, supplementary processing steps must follow to concatenate all the edges into edge chains that correspond better with borders in the image.

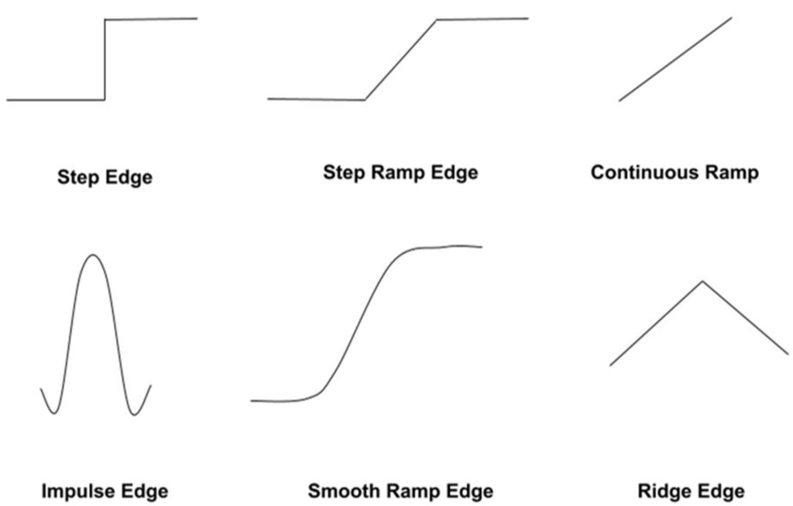
Image Segmentation: Methodologies

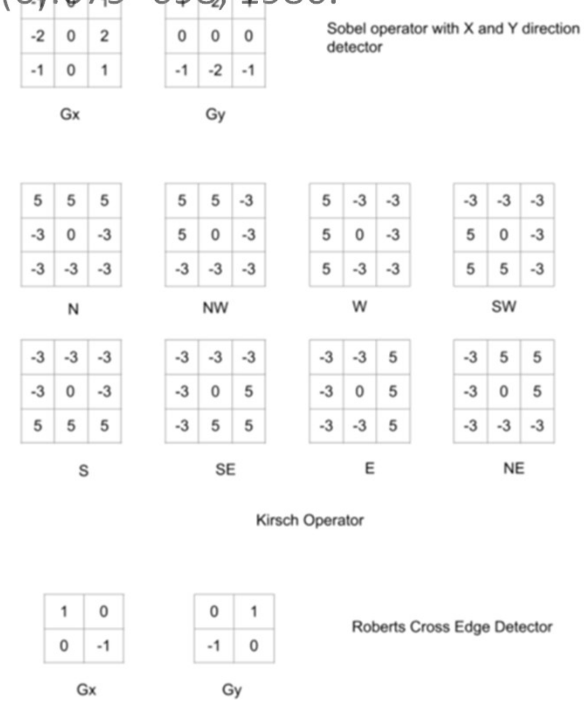
Here are different methods that one can employ to perform image segmentation:

2. Edge Based Methods (Boundary based approach)

   Type of edges and edge detectors

Canny method: John Canny. A computational approach to edge detection. Pattern Analysis and Machine Intelligence, IEEE Transactions on, (6):679–698, 1986.





$$g = \left(g_x^2 + g_y^2\right)$$

$$g = |g_x| + |g_y|$$

$$g = max(|g_x|, |g_y|)$$

$$\theta = arctan\frac{g_y}{g_x}$$

Image Segmentation: Methodologies

Here are different methods that one can employ to perform image segmentation:

2. Edge Based Methods (Boundary based approach)

   Canny method:

   John Canny. A computational approach to edge detection. Pattern Analysis and Machine Intelligence, IEEE Transactions on, (6):679–698, 1986.

   1. Apply Gaussian filter to smooth the image in order to remove the noise
   2. Find the intensity gradients of the image
   3. Apply gradient magnitude thresholding or lower bound cut-off suppression to get rid of spurious response to edge detection
   4. Apply double threshold to determine potential edges
   5. Track edge by hysteresis: Finalize the detection of edges by suppressing all the other edges that are weak and not connected to strong edges.

More: https://towardsdatascience.com/canny-edge-detection-step-by-step-in-python-computer-vision-b49c3a2d8123

Image Segmentation: Methodologies

Here are different methods that one can employ to perform image segmentation:

3. Region-Based Segmentation

   The region based segmentation methods involve the algorithm creating segments by dividing the image into various components having similar characteristics. These components, simply put, are nothing but a set of pixels. Region-based image segmentation techniques initially search for some seed points – either smaller parts or considerably bigger chunks in the input image.

   A. Region Growing: In this method, you start with a small set of pixels and then start iteratively merging more pixels according to particular similarity conditions. A region growing algorithm would pick an arbitrary seed pixel in the image, compare it with the neighbouring pixels and start increasing the region by finding matches to the seed point (bottom to up).

   A. Region Splitting and Merging: The splitting and merging based segmentation methods use two basic techniques done together in conjunction – region splitting and region merging – for segmenting an image. Splitting involves iteratively dividing an image into regions having similar characteristics and merging employs combining the adjacent regions that are somewhat similar to each other.

Image Segmentation: Methodologies

Here are different methods that one can employ to perform image segmentation:

3. Region-Based Segmentation: Region Growing

a. Choose a random pixels
b. Use 8-connected and threshold to determine
c. Repeat a and b until almost points are classified.

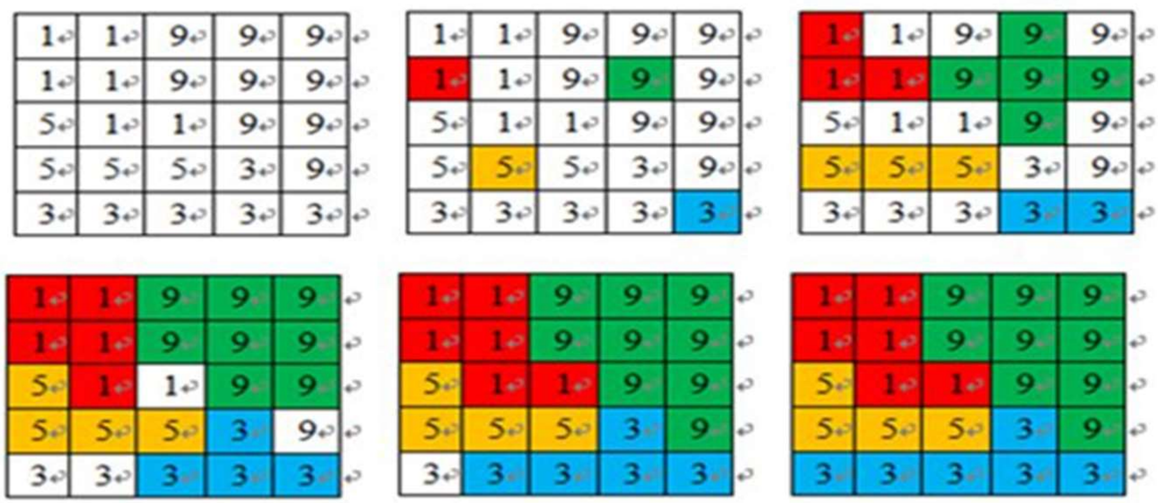


Image Segmentation: Methodologies

Here are different methods that one can employ to perform image segmentation:

3. Region-Based Segmentation: Region Splitting and Merging

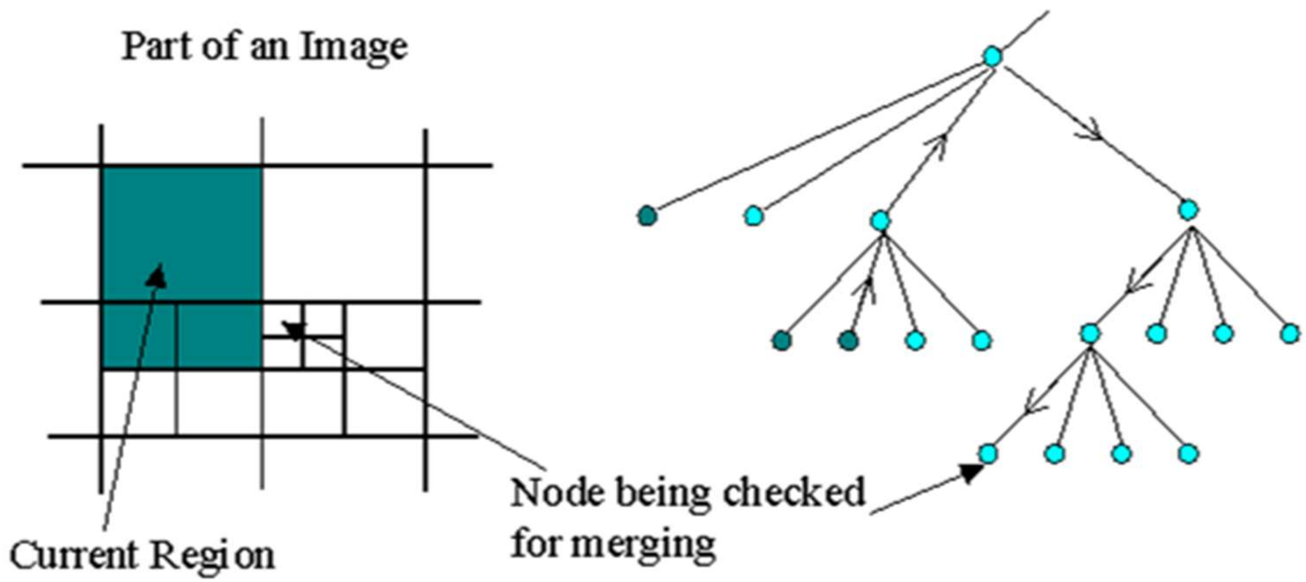


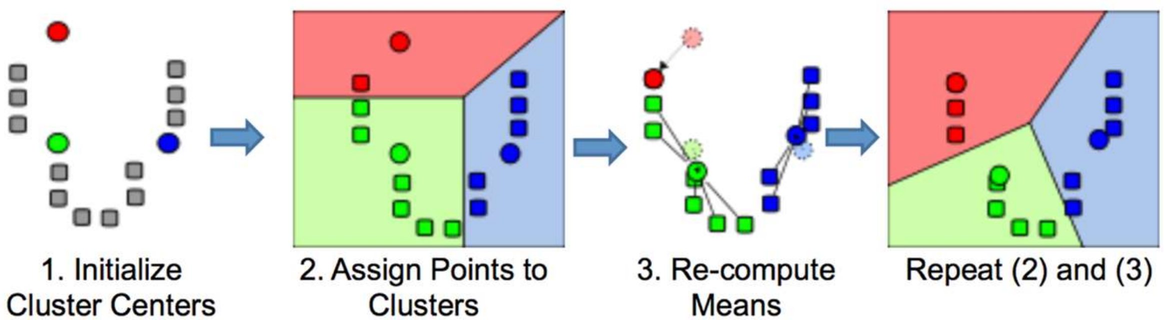
Image Segmentation: Methodologies

Here are different methods that one can employ to perform image segmentation:

4. Clustering Based methods:

   Clustering algorithms are unsupervised algorithms, unlike Classification algorithms, the user has no pre-defined set of features, classes, or groups. Clustering algorithms help in fetching the underlying, hidden information from the data like, structures, clusters, and groupings that are usually unknown from a heuristic point of view.

   1. K-Means



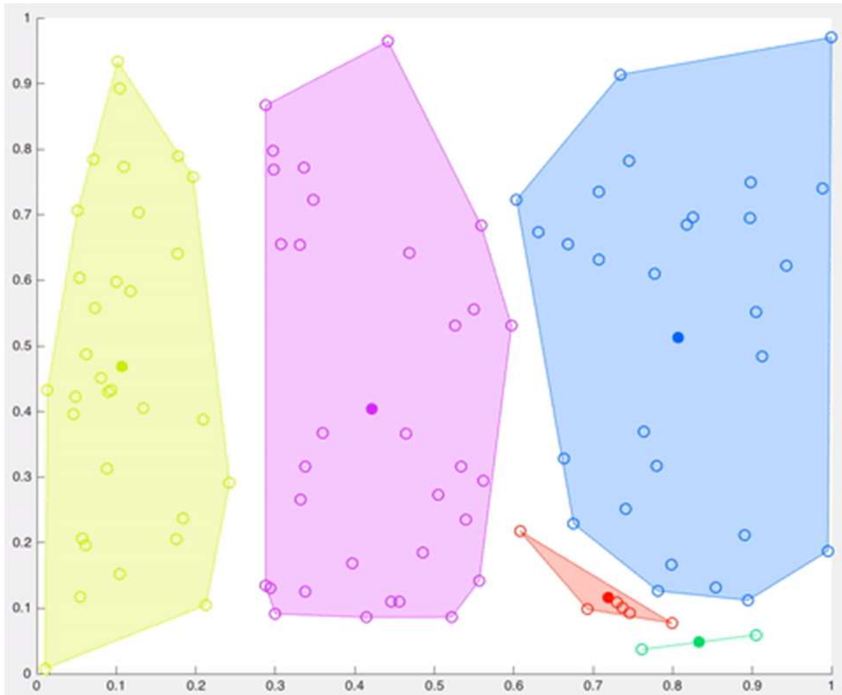


Image Segmentation: Methodologies

Here are different methods that one can employ to perform image segmentation:

4. Clustering Based methods:
   2. Hierarchical clustering (Bottom-up)

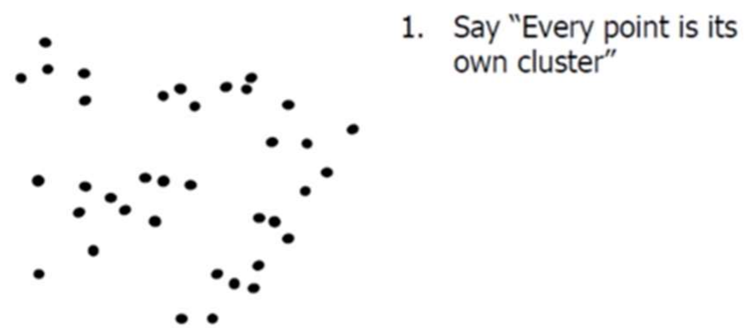


Image Segmentation: Methodologies

Here are different methods that one can employ to perform image segmentation:

4. Clustering Based methods:
    2. Hierarchical clustering (Bottom-up)

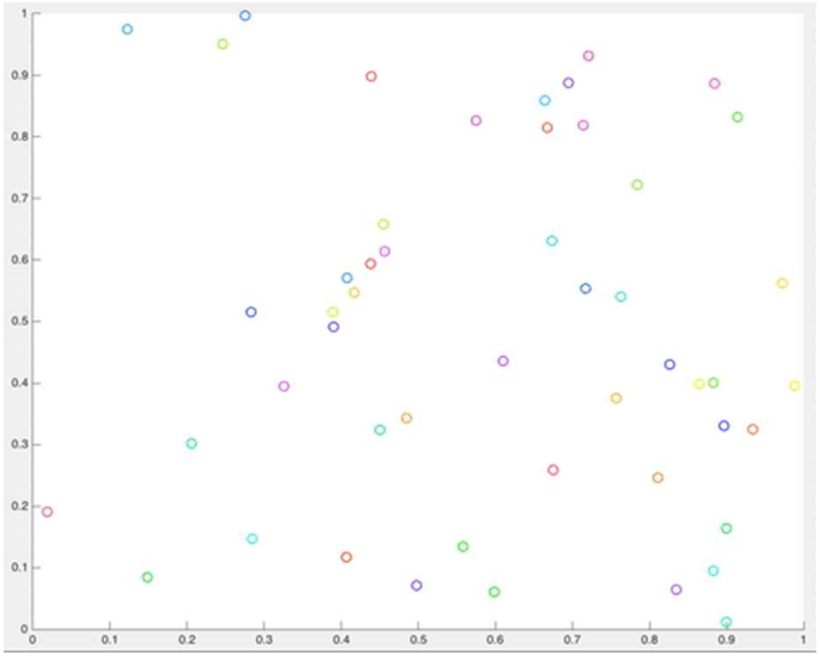


Image Segmentation: Methodologies

Here are different methods that one can employ to perform image segmentation:

5. Watershed based methods:

   In image processing, a watershed is a transformation on a grayscale image. It refers to the geological watershed or a drainage divide. A watershed algorithm (also a region-based method) would handle the image as if it was a topographic map. It considers the brightness of a pixel as its height and finds the lines that run along the top of those ridges. The slope and elevation of the said topography are distinctly quantified by the gray values of the respective pixels – called the gradient magnitude.

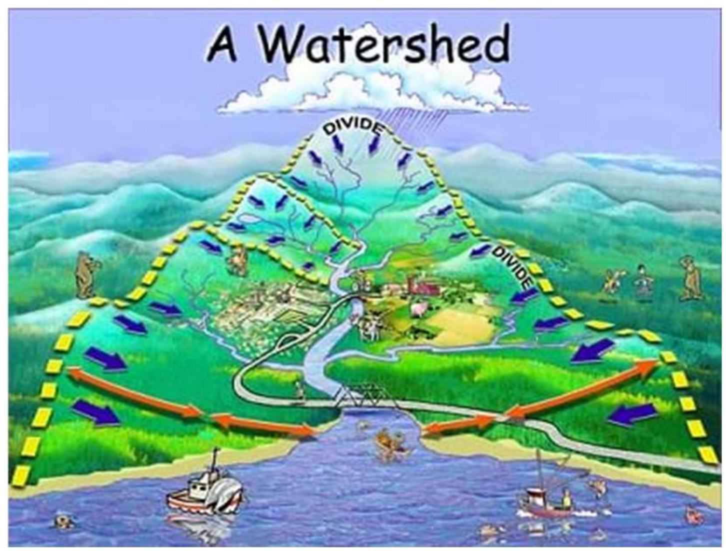


Image Segmentation: Methodologies

Here are different methods that one can employ to perform image segmentation:

6. Neural Network based methods:

   Another methodology to segment images is by using Neural Network. We will discuss this in more details

Image Segmentation: Methodologies

Techniques	Description	Advantages	Disadvantages
Thresholding Method	Focuses on finding peak values based on the histogram of the image to find similar pixels	Doesn't require complicated pre-processing, simple	Many details can get omitted, threshold errors are common
Edge Based Method	based on discontinuity detection unlike similarity detection	Good for images having better contrast between objects.	Not suitable for noisy images
Region-Based Method	based on partitioning an image into homogeneous regions	Works really well for images with a considerate amount of noise, can take user markers for fasted evaluation	Time and memory consuming
Traditional Segmentation Algorithms	Divides image into k number of homogenous, mutually exclusive clusters – hence obtaining objects	Proven methods, reinforced with fuzzy logic and more useful for real-time application.	Determining cost function for minimization can be difficult.
Watershed Method	based on topological interpretation of image boundaries	segments obtained are more stable, detected boundaries are distinct	Gradient calculation for ridges is complex.
Neural Networks	based on deep learning algorithms – Convolutional Neural Networks	easy implementation, no need for following any complicated algorithms, ready-made libraries available in Python, more practical applications	Training the model for custom and business images is time consuming and resource costly.

Python Practice 1: Image segmentation using these methods:

1. Threshold Method
2. Edge Based Segmentation
3. Region Based Segmentation
4. Clustering Based Segmentation
5. Watershed Based Method